

Yashveer Jain

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EDUCATION

Master of Engineering (Robotics) (GPA:3.85/4) University of Maryland – College Park (UMD), USA	Aug 2022–May 2024
Bachelor of Engineering (Mechatronics) (GPA:9.44/10) Manipal University Jaipur, Rajasthan, India	July 2016- July 2020

WORK EXPERIENCE

Embedded AI Intern - Renesas Electronics, USA - Trained and deployed SVM and logistic regression models on Renesas RA6 Series and RL78 microcontroller, achieving 95% accuracy for an Asset Tracking Project, enabling efficient asset management. - Ensured success of edge computing projects by developing and implementing machine learning models optimized for resource-constrained embedded systems.	May 2023–Aug 2023
Machine learning Engineer - AIMonk Labs, India - Developed and deployed a robust testing infrastructure for AI models, integrating FastAPI backend on AWS EC2 instance, streamlining the model development lifecycle. - Optimized OCR algorithms using Transformers implemented on PyTorch, achieving a 60% accuracy boost, significantly improving text recognition capabilities. - Enhanced car detection AI by 20% using DeepLabv3 and YOLOv7 integration, leading to more accurate object detection in autonomous systems. - Collaborated with cross-functional teams on Azure DevOps for model deployment on AWS SageMaker, ensuring seamless integration and deployment of machine learning models.	Aug 2020–Aug 2022
Research Intern - Swaayatt Robotics, India - Implemented Kalman and particle filters with LK optical flow using OpenCV for vehicle and pedestrian tracking in autonomous driving systems. - Implemented UnDeepVO paper unsupervised deep learning visual odometry paper using TensorFlow 2.0 and Keras for depth estimation and translation and rotation motion, increasing the SLAM accuracy by 20%.	Jan 2020–May 2020

RESEARCH AND TEACHING EXPERIENCE

Grader, University of Maryland (Spring 2024) ENPM808z – Cognitive Robotics with Prof. Yantian Zha, a graduate course at UMD. Adept at assessing student progress in integrating human cognition with robotics and AI, using simulators like NVIDIA Issac Sim, and Meta Habitat 2.0.	Jan 2024-May 2024
Graduate Research Assistant, University of Maryland (Fall 2022-Spring 2023) <i>TinyDepth: Generalized Neural Metric Depth for Palm-sized Robots. Under Review Nature npj Robotics.</i> Designed sub-6cm microdrone with embedded Raspberry Pi and ROS-based vision pipeline for autonomous planning and control (Perception & Robotics Group, Prof. Yiannis Aloimonos).	Aug 2022-May 2023

PROJECTS

SVM-based Image Classification with Multithreading (C++)- GitHub Implemented a Support Vector Machine (SVM) for image classification in C++. Utilized multithreading for improved performance on the CIFAR-10 dataset.	Mar 2024-May 2024
Home Automation - GitHub Established a home server for device connectivity using FastAPI, MQTT, Arduino, Raspberry Pi. Demonstrating proficiency in hardware integration and IoT systems.	Jun 2021-Aug 2021
Third eye - GitHub Built blind navigation device using Siamese Network (face recognition), Yolo9000 (object detection), and Text2Speech.	Jun 2021-Aug 2021
Search and Rescue Drone - GitHub Developed an AI Drone to detect human beings using RetinaNet for object detection in hazardous areas and send their locations to the Rescue team, showcasing practical application of AI in real-world scenarios.	Jul 2019-Oct 2019

SKILLS AND LANGUAGES

Machine Learning: TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV, Scikit-learn, Vector-c++, Huggingface.
Programming Languages: Python, C, C++
Development Tools: Git, Docker, Agile Development
Cloud Platforms: AWS (Certified Cloud Practitioner), Azure
Robotics: ROS (Robot Operating System), ROS2, Arduino, Raspberry Pi, Nvidia Jetson.

CERTIFICATE

Building Transformer-Based Natural Language Processing Applications (Nvidia) Training and Deploying BERT Transformer Model on NER and Text classification and deployed on Triton Server.	Feb 2024
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